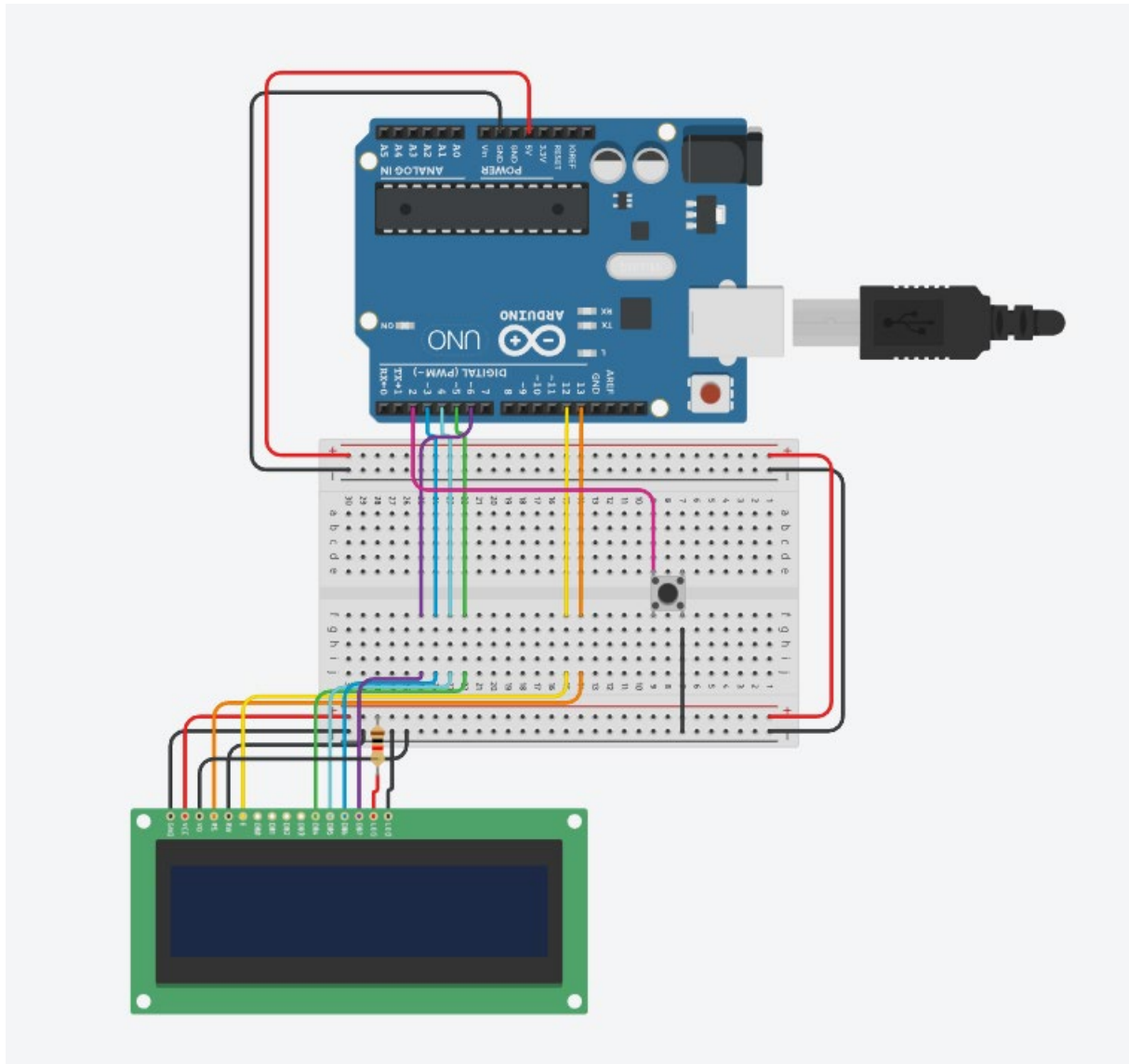




```
#include <LiquidCrystal.h>
#include <avr/interrupt.h>

LiquidCrystal lcd(13, 12, 5, 4, 3, 6);

const int buttonPin = 2;

volatile int tenths = 0;
volatile int seconds = 0;
volatile int minutes = 0;

enum TimerState { STOPPED, RUNNING };
volatile TimerState timerState = STOPPED;

unsigned long buttonPressStart = 0;
bool buttonHeld = false;

void setup() {
  Serial.begin(9600);
```

```

lcd.begin(16, 2);
lcd.setCursor(0, 0);
lcd.print("Stopwatch Timer");
lcd.setCursor(0, 1);
lcd.print("00:00:0");

pinMode(buttonPin, INPUT_PULLUP);
attachInterrupt(digitalPinToInterrupt(buttonPin), handleButtonPress, FALLING);

cli();
TCCR1A = 0;
TCCR1B = 0;
TCNT1 = 0;
OCR1A = 1562.5;
TCCR1B |= (1 << WGM12);
TCCR1B |= (1 << CS12) | (1 << CS10);
TIMSK1 |= (1 << OCIE1A);
sei();
}

ISR(TIMER1_COMPA_vect) {
  if (timerState == RUNNING) {
    tenths++;
    if (tenths >= 10) {
      tenths = 0;
      seconds++;
      if (seconds >= 60) {
        seconds = 0;
        minutes++;
      }
    }
  }
  updateDisplay();
}

void updateDisplay() {
  lcd.setCursor(0, 1);
  lcd.print((minutes < 10 ? "0" : "") + String(minutes) + ":");
  lcd.print((seconds < 10 ? "0" : "") + String(seconds) + ":");
  lcd.print(String(tenths));
}

void handleButtonPress() {
  buttonPressStart = millis();
  buttonHeld = true;
}

void checkButtonHold() {
  if (buttonHeld && digitalRead(buttonPin) == LOW) {
    if (millis() - buttonPressStart > 3000) {
      timerState = STOPPED;
      tenths = 0;
      seconds = 0;
      minutes = 0;
      updateDisplay();
      buttonHeld = false;
    }
  } else if (buttonHeld && digitalRead(buttonPin) == HIGH) {

```

```
buttonHeld = false;
if (timerState == STOPPED){
    timerState = RUNNING;
}else {
    timerState = STOPPED;
}
}
}

void loop() {
    checkButtonHold();
}
```